

Project Submission — Start Here

Predicting Diabetes Risk from Behavioural and Demographic Health Indicators

EE-439 — Introduction to Machine Learning

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1. What this project does

Type 2 diabetes is common, slow to develop and frequently undiagnosed. Confirming it requires a laboratory blood test, which is not affordable at population scale. This project asks whether a machine learning model can identify people at high risk from self-reported health and lifestyle information alone — answers anyone can give in a few minutes with no medical equipment — so that limited testing capacity can be pointed at the people most likely to need it.

It is a supervised binary classification problem, built on 253,680 responses to the CDC's 2015 Behavioral Risk Factor Surveillance System survey.

2. Headline result

	Result
Best model	LightGBM (gradient-boosted decision trees)
Test ROC-AUC	0.814
Diabetic cases identified	77.3% (6,140 of 7,945)
Evaluated on	45,957 held-out records, opened once

Four models were trained and compared: Logistic Regression, Decision Tree, Gaussian Naive Bayes and LightGBM. Section 9 of the report explains the choice, and Section 8 compares all four models side by side.

3. The pipeline, and where each stage is documented

Stage	Report section
Data collection	2. Dataset
Data preprocessing and cleaning	3. Data Preprocessing and Data Cleaning
Feature extraction and selection	4. Feature Extraction and Selection
Model selection	5. Model Selection
Training	6. Training
Validation	7. Validation and Results
Best model and reasoning	8. Selection of the Best Model
Testing and final results	9. Testing: Final Evaluation on the Held-Out Set

4. Where to find things

Read the report first — everything else is supporting material.

Folder	Contents
01_Report	The full report (46 pages), as PDF and as Word. This is the main deliverable. Eleven sections plus Appendix A, which reproduces the complete source code.
02_Code	Three Jupyter notebooks, numbered in the order they should be read. All outputs are saved, so they can be read without being run. The same code appears in Appendix A of the report.
03_Dataset	The dataset CSV, plus a note recording its source and licence.
04_Results	Everything the code produced: 13 figures, 6 trained models, and the result tables.
05_Environment	How the project was built and how to reproduce it.

5. The three notebooks

Each notebook writes files the next one reads, so run them in order.

Notebook	What it does	Report §
01_data_exploration_and_preprocessing.ipynb	Loads and explores the data, then cleans it: removes duplicates, clips BMI outliers, converts the target to binary, scales features, makes the train/test split.	2–3
02_model_training_and_selection.ipynb	Trains all four models and compares them by five-fold cross-validation on the training set only.	4–7
03_final_evaluation.ipynb	Opens the held-out test set once and evaluates the chosen model.	8–9

6. Running the code

Full instructions are in 05_Environment. The short version:

```
cd 02_Code
pip install -r requirements.txt
jupyter notebook          # or open the folder in VS Code
```

The notebooks expect the dataset and intermediate files in the same folder as the notebook. If you run them, copy the CSV from 03_Dataset into 02_Code first, or adjust the path in the loading cell of notebook 01.

7. One number that looks wrong but isn't

The models score 69.7%–71.5% accuracy, while answering “no diabetes” to everybody would score 82.71%. That is not a failure. Only 17.29% of respondents are diabetic, so accuracy rewards ignoring them. The models were deliberately weighted to stop doing that, trading raw accuracy for the ability to find roughly

three quarters of the diabetic cases instead of none. Section 7.2 of the report sets this out, and balanced accuracy is reported alongside.